Beyond Short Names: Cross-Model Emoji Lexical Neighborhoods Without a Simple Late-Layer Mediator

Anonymous Author(s)
Affiliation
Address
email

Abstract

Emoji reach text language models as Unicode token sequences, not rendered pictures. We ask whether a raw atomic emoji indexes only its official short name or a wider lexical neighborhood, and whether that relation has a simple late-layer mediator. We prospectively cross 55 emoji and 55 non-emoji Unicode symbols with raw, canonical-name, lexically disjoint CLDR-keyword, and literal codepoint views. Across nine frozen model/tokenizer cells, the emoji alias-minus-codepoint retrieval effect is positive in both unrestricted and exact four-view token-count-matched analyses. All nine cells exceed a 0.02 practical floor; the registered at-least-three-of-four core-model primary passes with four positive cells and partial-conjunction $p = 0.0027$. Canonical names are also retrieved for symbols, but symbol alias effects are smaller and heterogeneous in this registered secondary comparison. Passing the observational gate admits a separate causal test: rank-8 subspaces learned on 128 source emoji are intervened on for 192 untouched emoji at six late attention and MLP outputs in Llama 3.2 3B. Engineering controls are exact, yet no primary site is jointly necessary and sufficient. The combined result is a controlled dissociation: a repeatable text-only emoji-to-CLDR lexical relation exists, while the preregistered sparse late-component account does not explain it.

1 Introduction

Emoji are compact visual conventions for people, but a text language model receives them as encoded Unicode scalars and tokenizer-dependent sequences. That distinction matters for interpretation. A representation probe may recover an official label because the label co-occurs with the character in standards, documentation, or training data. Such a result need not imply rendered-glyph perception, a human-like concept, or a localized semantic circuit.

We therefore ask a narrower two-part question. First, does a raw atomic emoji retrieve a lexically disjoint CLDR keyword better than its literal `U+...` spelling after exact token-count matching? Second, if this observational relation is reliable, can a frozen low-rank late-component intervention identify a representation that is both necessary and sufficient for it? The intended contribution is not a claim that models "understand emoji." It is an auditable separation between a broad lexical readout and one simple mechanistic account.

Our design contributes three pieces of evidence:

- A four-view, three-bank protocol separates raw Unicode surface, official English CLDR name, a word-disjoint CLDR keyword, and literal codepoint spelling. The causal source and evaluation banks are untouched by the observational run.

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- The preregistered observational primary passes: 4/4 core cells and 9/9 total cells are positive in unrestricted and exact four-view token-count matched analyses. The cross-model partial-conjunction value is 0.0027.

- A conditionally admitted causal experiment passes all zero-hook and restoration controls but finds no primary block-24 attention or MLP output that is jointly necessary and sufficient. A secondary block-27 MLP patch is sufficient-only and is not a complete mediator.

The result rules out a narrow exact-short-name account under the frozen endpoint and rejects one preregistered rank-8 late-component mechanism. It does not identify which of distributed, redundant, nonlinear, or item-private alternatives is correct.

2 Related work

Emoji as lexical objects. Prior work learns emoji embeddings from descriptions [Eisner et al., 2016] and builds sense inventories from lexical and usage resources [Wijeratne et al., 2016]. We instead treat CLDR names and keywords [Unicode Consortium, 2026] as fixed, source-auditable endpoints for probing frozen hidden states. The goal is not to produce a better emoji embedding, but to test which textual relation is recoverable from a raw Unicode input and which alternative explanations remain.

Probing and tokenizer controls. Probe accuracy can reflect properties of the control task rather than the target representation [Hewitt and Liang, 2019]; tokenization itself can materially change language-model comparisons [Rust et al., 2021]. We avoid a learned diagnostic classifier, use item-wise retrieval, and require both unrestricted and exact four-view token-count-matched tests. This is a token-count control, not token-identity matching or tokenizer independence. Our readout is also more local than whole-representation similarity analyses such as CKA [Kornblith et al., 2019]: it asks whether the correct item is the nearest cross-view partner.

Causal intervention. Causal mediation and activation interventions can distinguish readable correlates from components that affect model computation [Vig et al., 2020, Meng et al., 2022, Turner et al., 2023]. We adopt that distinction but require a frozen site to pass both random-adjusted necessity and sufficiency on a disjoint evaluation bank. The negative result is retained rather than followed by an adaptive search over layers, ranks, or endpoints.

3 Prospective design

3.1 Claim boundary, views, and banks

Throughout, *glyph* means the raw Unicode scalar string supplied to a text model. No raster image is rendered. Atomic emoji are single-codepoint, fully-qualified Emoji 17 entries introduced no later than Emoji 13.1 [Unicode Consortium, 2025]. For each item, the canonical view is the exact English CLDR 48.2 `tts` value; the alias is one exact English CLDR `default` keyword. After lowercasing and extracting ASCII `[a-z0-9]+` words, canonical and alias word sets must be nonempty and disjoint. The fourth view is the uppercase literal codepoint spelling.

The observational bank contains 55 emoji and 55 non-emoji Unicode symbols. Symbols are single scalars in a Unicode general category beginning with `S`, have `Emoji=false`, have valid disjoint CLDR aliases, and are absent from every strict-UTF-8, no-NUL repository blob at the frozen boundary. This is repository-relative freshness, not pretraining novelty. Exact emoji surfaces may have appeared in earlier experiments; the endpoint, alias assignment, and bank split are prospective, not the emoji surfaces.

Two further banks contain 128 causal-source and 192 causal-evaluation emoji. All 430 codepoints and 430 aliases are unique across banks. Figure 1 summarizes the separation and the one-way admission gate.

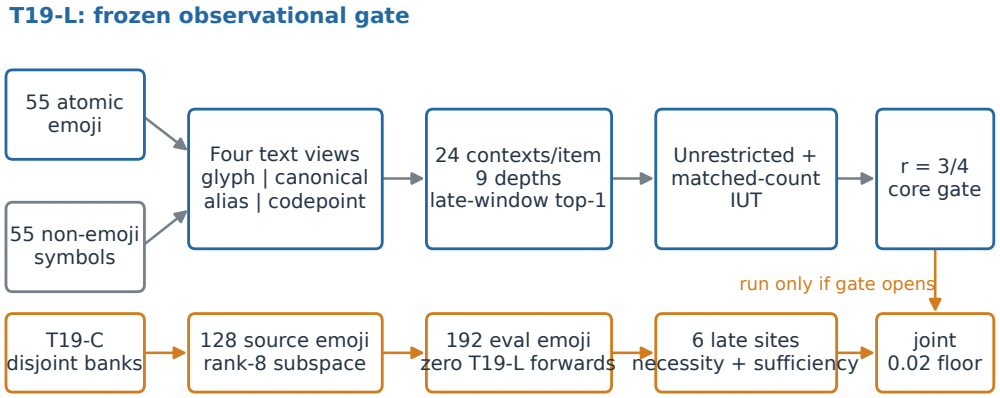


Figure 1: Prospective T19 design. T19-L crosses two glyph kinds with four text views. Only the emoji endpoint enters the registered cross-model primary. T19-C runs only after that gate opens and uses disjoint source and evaluation emoji. No causal-bank item receives a T19-L forward.

3.2 Observational readout

For each item and view, the frozen harness appends one of two suffix carriers (`" a"` or `" |"`) at distances 1, 2, 4, and 8 and records the hidden state at the selected suffix token at nine normalized depths. Batch size is one; no chat template or added special token is used. The primary late window is fixed at depths 0.75, 0.875, and 1.0, yielding 24 contexts per item.

Within each view and context, hidden states are centered across items and each row is L2-normalized. Let $z_i^{v,c}$ denote the resulting vector for item i, view v, and context c. Cross-view top-1 retrieval is

$$R_c(g \to v) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\left[i = \arg\max_j \left\langle z_i^{g,c}, z_j^{v,c} \right\rangle \right]. \tag{1}$$

The per-model primary estimate is

$$\Delta_m = \frac{1}{24} \sum_{c} \left[R_c(g \to \textsc{alias}) - R_c(g \to \textsc{codepoint}) \right] \tag{2}$$

on the 55 observational emoji. This is a nearest-neighbor readout from hidden states, not a next-token generation score.

3.3 Token-count intersection and cross-model test

Every model must pass two components. The unrestricted component uses all 55 emoji and permutes alias correspondence within glyph kind. The matched component retains non-singleton strata sharing the exact four-view token-count signature and permutes aliases only within stratum. Canonical and codepoint correspondences stay fixed. Each component uses 19,999 one-sided randomization draws; the model intersection-union p-value is the maximum of the two component values. Both observed effects must also be at least 0.02.

The sole cross-model primary uses four frozen core cells: Llama 3.2 3B base, Qwen2.5 7B base, OLMo 2 1B base, and Gemma 2 2B IT. It tests the partial conjunction that at least $r = 3$ of $m = 4$ cells pass [Benjamini and Heller, 2008]. With ordered model IUT values $p_{(1)} \leq \cdots \leq p_{(4)}$, the registered value is $p_{\mathrm{PC}} = \min(1, 2p_{(3)})$. The other five cells and all symbol or canonical-name results are secondary and cannot open the causal gate.

We report percentile intervals from 20,000 item bootstrap replicates as descriptive uncertainty. They are not substituted for the permutation-based primary decision.

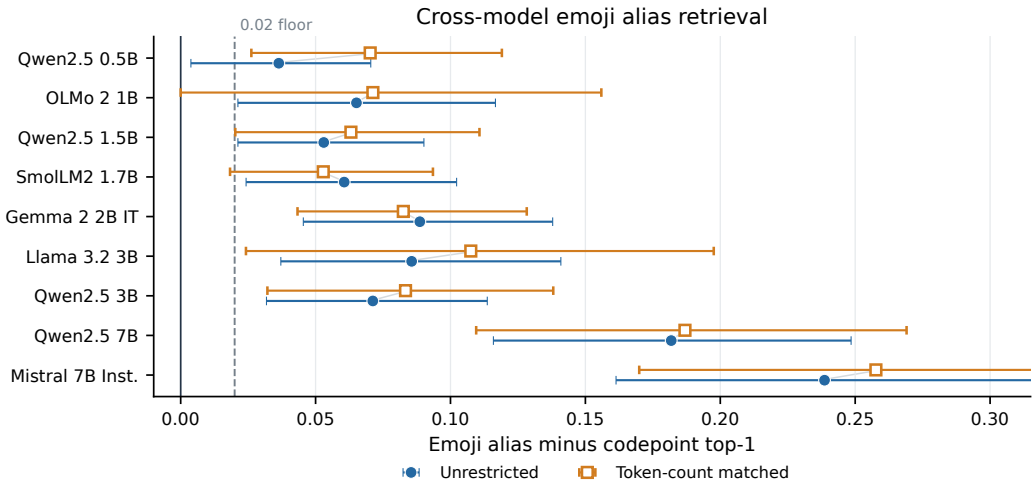


Figure 2: Emoji alias-minus-codepoint top-1 effects in the registered roster order. Bars are descriptive 95% item-bootstrap intervals (20,000 replicates); the randomization tests determine the primary decision. Every point estimate exceeds the preregistered 0.02 floor in unrestricted and exact token-count-matched analysis.

3.4 Conditionally admitted causal test

After the observational gate was committed, T19-C ran in the frozen Llama 3.2 3B base cell. At each of six sites (attention and MLP component outputs at blocks 21, 24, and 27), we learned a rank-8 basis from the left singular directions of the centered, row-normalized glyph-alias cross-covariance on 128 source emoji. A deterministic rank-8 control basis was drawn from the same source span and made orthogonal to the learned basis. Blocks 24 attention and MLP were primary; blocks 21 and 27 were secondary.

On 192 different evaluation emoji, necessity removes learned coordinates from the glyph component output. Sufficiency adds the same-item learned glyph-minus-codepoint coordinates to the codepoint output. Analogous operations in the random basis define adjusted effects:

$$N_s = E_s(\text{random removal}) - E_s(\text{learned removal}), \tag{3}$$

$$S_s = E_s(\text{learned patch}) - E_s(\text{random patch}), \tag{4}$$

where E_s is the final-hidden alias-minus-codepoint top-1 endpoint. A site must have both effects and both lower 95% item-bootstrap bounds above 0.02, pass both one-sided sign-flip tests (99,999 draws), pass zero and remove-restore engineering controls, and survive Holm correction across the two primary sites [Holm, 1979]. No site, rank, depth, carrier, subset, or endpoint may be replaced after seeing the result.

4 Results

4.1 A lexical relation beyond the official short name

The registered cross-model primary passed. All four core cells were positive; the third-smallest core IUT value was 0.00135, yielding $p_{\text{PC}} = 2(0.00135) = 0.0027$. All nine frozen cells were positive and exceeded the 0.02 practical floor in both unrestricted and matched analyses (Figure 2; exact values in Appendix B). Model IUT values ranged from 0.00005 to 0.00185. An independent implementation reconstructed all 36 glyph-kind by component randomization tests and the same gate decision.

Across models, mean emoji alias effects were +0.0979 unrestricted and +0.1084 matched. Mean canonical-name effects were larger, +0.1440 and +0.1718. Thus, the result is wider than exact

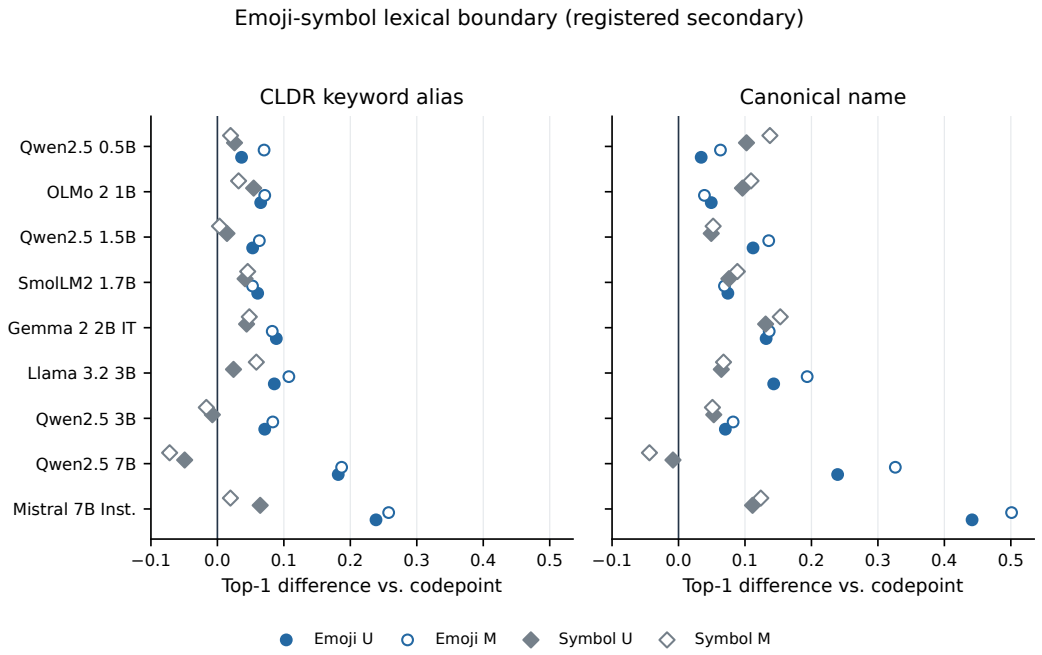


Figure 3: Registered secondary comparison. Each row shows unrestricted (U) and token-count-matched (M) effects for the same frozen model. Official-name retrieval is not exclusive to emoji, while disjoint-keyword effects are stronger and more consistent for the frozen emoji panel. Glyph category, frequency, label quality, and token identity are not exchangeable, so this is a boundary result rather than an emoji-only semantic claim.

official-name retrieval, but the official name remains the stronger key on average. The alternative label is an exact CLDR keyword, not a human-rated synonym or paraphrase; co-occurrence in model training remains a sufficient origin for the relation.

Depth was secondary. The unrestricted emoji alias contrast averaged $+0.0409$ at the first block, peaked at $+0.1152$ at normalized depth 0.875, and ended at $+0.0922$. Individual cells had different onset and peak depths, so these data do not support one universal depth trajectory.

4.2 Named symbols bound emoji specificity

Non-emoji symbols showed mean alias effects of $+0.0236$ unrestricted and $+0.0153$ matched, compared with $+0.0979$ and $+0.1084$ for emoji. The emoji-minus-symbol point estimate was positive in all nine cells in both components. Yet symbol aliases were heterogeneous: Qwen2.5 3B and 7B were negative in both components. Canonical-name retrieval exceeded alias retrieval for symbols in 9/9 cells in both components, with mean symbol canonical effects of $+0.0750$ and $+0.0821$ (Figure 3).

This comparison cannot establish an emoji-specific faculty. The panels are not matched on training frequency, lexical specificity, Unicode category, visual familiarity, or token identity. Its role is to prevent the stronger but false inference that only emoji acquire readable names.

4.3 The simple late-component mechanism fails

All six causal sites passed their engineering checks: zero hooks and remove-restore conditions reproduced the baseline final hidden state with exactly zero RMS difference. Nevertheless, neither primary block-24 site passed. Block-24 attention had adjusted necessity 0.0000 and sufficiency -0.00521; block-24 MLP had $+0.00521$ and 0.0000. Their site IUT values were 1.0 and 0.74912, respectively. No one of the six sites passed both components (Figure 4; exact values in Appendix C).

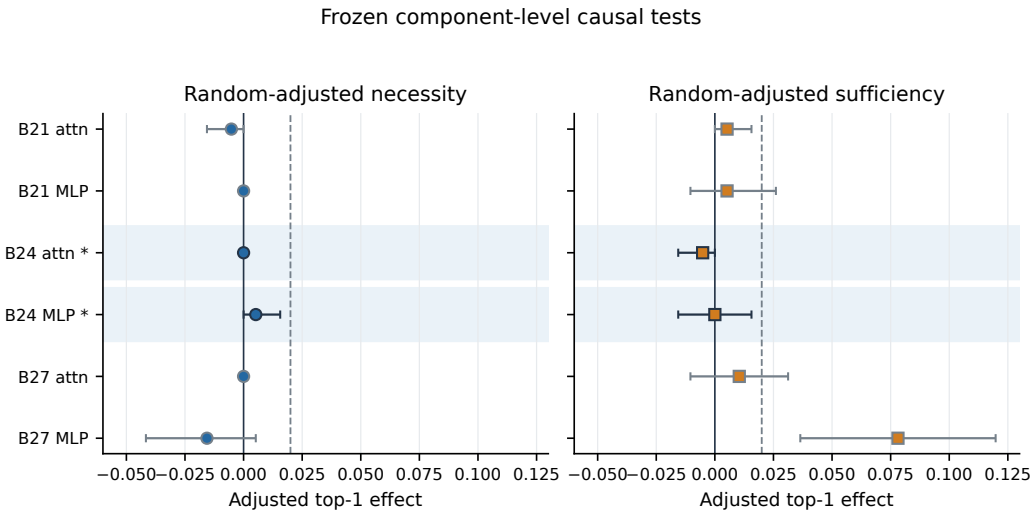


Figure 4: Random-adjusted causal effects with descriptive 95% item-bootstrap intervals over 192 untouched emoji. Shaded rows and asterisks mark the two primary block-24 sites. Solid vertical lines show zero; dashed lines show the 0.02 practical floor. No site has both necessity and sufficiency above the floor with positive lower bounds.

The secondary block-27 MLP patch produced adjusted sufficiency $+0.07812$ (95% interval $[+0.03646, +0.11979]$, one-sided $p = 0.00009$), but necessity was -0.01562 (interval $[-0.04167, +0.00521]$, $p = 0.96895$). The site IUT was therefore 0.96895. Moreover, the learned patch raised alias top-1 from 0.0052 to 0.0208 while lowering codepoint self-retrieval from 1.0000 to 0.9375. It is an incomplete perturbation effect, not clean alias insertion or a necessary mechanism.

5 Discussion

The observational and causal results answer different questions. T19-L shows that, under a fixed text-only readout, the same item is recoverable across raw emoji and a lexically disjoint CLDR keyword more reliably than across raw emoji and a literal codepoint string. Exact token-count matching makes a pure length account insufficient. The symbol panel shows that official-name readout is a broader Unicode phenomenon, while keyword breadth is graded rather than categorically emoji-only.

T19-C then rejects a tempting compression of that result: a shared rank-8 late attention or MLP component, learned from one bank and tested on another, is not jointly necessary and sufficient at the frozen sites. This does not turn the observational result into an artifact. It says that aggregate linear readability does not entail one sparse causal mediator. Distributed, redundant, nonlinear, and item-private accounts remain possible, but the present data do not distinguish them.

The block-27 MLP asymmetry gives a useful design lesson. A patch can move an endpoint without identifying the normal route that produces it. Requiring necessity, sufficiency, random-subspace adjustment, restoration, and a disjoint evaluation bank prevents that isolated patch success from being upgraded to a circuit claim.

6 Limitations and broader impact

Measurement. CLDR keywords are standardized annotations, not human judgments of synonymy, natural naming, specificity, or familiarity. Retrieval is nearest-neighbor geometry at selected suffix tokens, not generation or task behavior. Inputs are Unicode text; no rendered glyph is perceived. Exact token-count matching does not match token identities or equalize tokenizer construction.

Generalization. The nine model/tokenizer cells are a fixed, convenience-constrained roster spanning several families and sizes [Yang et al., 2024, Groeneveld et al., 2024, Dubey et al., 2024, Gemma Team et al., 2024, Jiang et al., 2023, Hugging FaceTB, 2024]. Model family, parameter count, training data, tokenizer, and instruction tuning are confounded, so the study supports neither a scale law nor a model-population estimate. Exact emoji surfaces are not pretraining-novel. The symbol panel is repository-text-fresh but not a pretraining holdout and is not exchangeable with the emoji panel.

Mechanism. The negative causal result covers one Llama 3.2 3B checkpoint, rank 8, one carrier and distance, two primary and four secondary sites, and one final-hidden endpoint. It does not show that no causal representation exists. A future study should preregister distributed multi-site removal, nonlinear or sparse readouts, and a new source/evaluation split rather than selecting a method on the present evaluation bank.

Reproducibility and impact. The compact archive binds configurations, selections, summaries, tests, and independent validators, but the current public package omits model-dependent raw arrays retained in execution scratch. An anonymized submission artifact and complete license/compute audit remain preparation items. Better Unicode auditing could reveal hidden representation failures in multilingual or social-media systems. Conversely, treating CLDR correspondence as a person's meaning, sentiment, culture, or identity could erase contextual variation and enable unjustified profiling. The measurements should not be used for such inference. No personal data, human subjects, or user messages are used.

7 Conclusion

Across nine frozen model/tokenizer cells, raw atomic emoji retrieve word-disjoint CLDR keyword aliases better than literal codepoints after exact four-view token-count matching. A named-symbol control bounds the specificity of that observation, and a separately banked causal experiment finds no primary rank-8 late component that is jointly necessary and sufficient. The scientific contribution is this dissociation: broad lexical readability is repeatable, but it should not be mistaken for visual understanding or a simple localized mechanism.

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A Frozen model roster

Table 1 reports the exact model artifacts and revisions used by the frozen observational roster. The Llama role uses the pinned BF16 mirror artifact specified by the protocol. Model cards and upstream license terms are part of the pending submission-archive license audit.

Role	Artifact identifier	Revision
Qwen2.5 0.5B base	`Qwen/Qwen2.5-0.5B`	`060db6499f32`
OLMo 2 1B base	`allenai/OLMo-2-0425-1B`	`a1847dff3500`
Qwen2.5 1.5B base	`Qwen/Qwen2.5-1.5B`	`8faed761d45a`
SmolLM2 1.7B base	`HuggingFaceTB/SmolLM2-1.7B`	`effd688a1292`
Gemma 2 2B IT	`google/gemma-2-2b-it`	`299a8560bedf`
Llama 3.2 3B base	`mlx-community/Llama-3.2-3B-bf16`	`60a99aaf4316`
Qwen2.5 3B base	`Qwen/Qwen2.5-3B`	`3aab1f1954e9`
Qwen2.5 7B base	`Qwen/Qwen2.5-7B`	`d14972939875`
Mistral 7B Inst. v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	`c170c708c41d`

Table 1: Frozen artifact identifiers and the first 12 hexadecimal characters of their complete pinned revisions. Full revisions are stored in the submission manifest.

B Exact observational results

Frozen cell	U effect	M effect	Model IUT p	Pass
Qwen2.5 0.5B	0.0364	0.0702	0.00055	yes
OLMo 2 1B	0.0652	0.0712	0.00135	yes
Qwen2.5 1.5B	0.0530	0.0631	0.00185	yes
SmolLM2 1.7B	0.0606	0.0528	0.00005	yes
Gemma 2 2B IT	0.0886	0.0825	0.00005	yes
Llama 3.2 3B	0.0856	0.1075	0.00135	yes
Qwen2.5 3B	0.0712	0.0833	0.00005	yes
Qwen2.5 7B	0.1818	0.1869	0.00005	yes
Mistral 7B Inst.	0.2386	0.2577	0.00005	yes

Table 2: Exact first-draft reporting values for the registered emoji alias-minus-codepoint endpoint. U is unrestricted; M is exact four-view token-count matched. The model IUT is the maximum component randomization p-value.

Matched emoji counts ranged from 31 to 41; tokenizer-only preflight retained at least 63 observational items total in every cell, with at least 31 emoji and 32 symbols. The acquisition comprised 7,920 forwards, 285,120 readout vectors, and 45 sealed shards.

C Exact causal results

Block	Site	Primary	N_s	p_N	S_s	p_S	Site IUT
21	attention	no	-0.00521	1.00000	0.00521	0.49983	1.00000
21	MLP	no	0.00000	1.00000	0.00521	0.49760	1.00000
24	attention	yes	0.00000	1.00000	-0.00521	1.00000	1.00000
24	MLP	yes	0.00521	0.49989	0.00000	0.74912	0.74912
27	attention	no	0.00000	1.00000	0.01042	0.31511	1.00000
27	MLP	no	-0.01562	0.96895	0.07812	0.00009	0.96895

Table 3: Random-adjusted necessity (N_s) and sufficiency (S_s). All six engineering gates passed. Holm correction applies across the two primary sites; no site passed the component-level intersection.

The causal run used 256 source forwards, 2,880 evaluation forwards, and 8,064 intervention rows. Each of 12 component tests used 99,999 sign flips and 20,000 item-bootstrap replicates. An independent validator rehashed source and evaluation arrays, reconstructed all 12 tests and intervals, verified learned and random subspace relations, and reproduced six engineering passes and zero positive sites.

D Execution and artifact boundary

Both observational and causal acquisition were frozen to a Transformers/CUDA BF16 backend and an NVIDIA GeForce RTX 5090. Model artifacts, tokenizer replay receipts, selection files, analysis summaries, randomization seeds, bootstrap seeds, and validators are checksum-bound. Figure inputs in this draft are only the compact validated model-cell and causal-site JSON summaries; a generated manifest hashes both inputs and every plotted PDF/PNG output.

The current compact Git artifact does not contain the model-dependent raw activation arrays or the full intervention ledger. Those remain bound by run receipts in execution scratch. Consequently, the draft supports audit of the published summary chain but does not yet provide a self-contained anonymous raw-data reproduction package. Exact wall-clock and peak-memory reporting is also not present in the compact T19 receipts. These are explicit submission gaps rather than inferred values.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Sections 1, 5, and 6 separate the positive observational result, the negative registered causal result, and their bounded interpretation.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 discusses measurement, tokenization, model-roster, causal-scope, archive, and social-interpretation limits.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper reports empirical tests and does not introduce a theoretical result.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [No]

   Justification: Sections 3 and A–D disclose the protocol and frozen revisions, but the current anonymous package and raw-array archive are not yet complete.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: A documented public development repository exists, but a double-blind submission artifact with the complete retained evidence has not yet been packaged.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

Justification: Section 3 specifies all evaluation banks, views, contexts, readouts, tests, gates, interventions, and fixed sites; no model training is performed.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Sections 3.3–3.4 define the randomization, partial-conjunction, sign-flip, Holm, and bootstrap procedures; Figures 2 and 4 label bootstrap intervals as descriptive.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: Appendix D identifies the RTX 5090 and BF16 CUDA path and reports forward counts, but exact wall-clock and peak-memory receipts are not in the compact archive.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [N/A]

   Justification: Formal author review against the current NeurIPS Code of Ethics remains a pre-submission checklist item.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Section 6 discusses auditing benefits and the risk of converting standardized lexical correspondence into unjustified cultural, sentiment, or identity inference.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The work releases no new pretrained model, scraped dataset, or high-risk capability; it studies public Unicode annotations and existing model checkpoints.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [No]

    Justification: Upstream resources and model reports are cited, but the complete per-model and CLDR/Unicode license audit is not yet included in the anonymous submission package.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [No]

    Justification: The protocol, selections, code, and compact evidence are documented in the development repository, but their final anonymous submission packaging is incomplete.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study uses no crowdsourcing or human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The study uses no crowdsourcing, personal data, or human subjects.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: Language models are the experimental objects and are fully identified; no LLM is an original or non-standard component used to design, label, or score the core experiment.